Tech Eximius 2.0 · Healthcare & MedTech · AI

MediSaathi

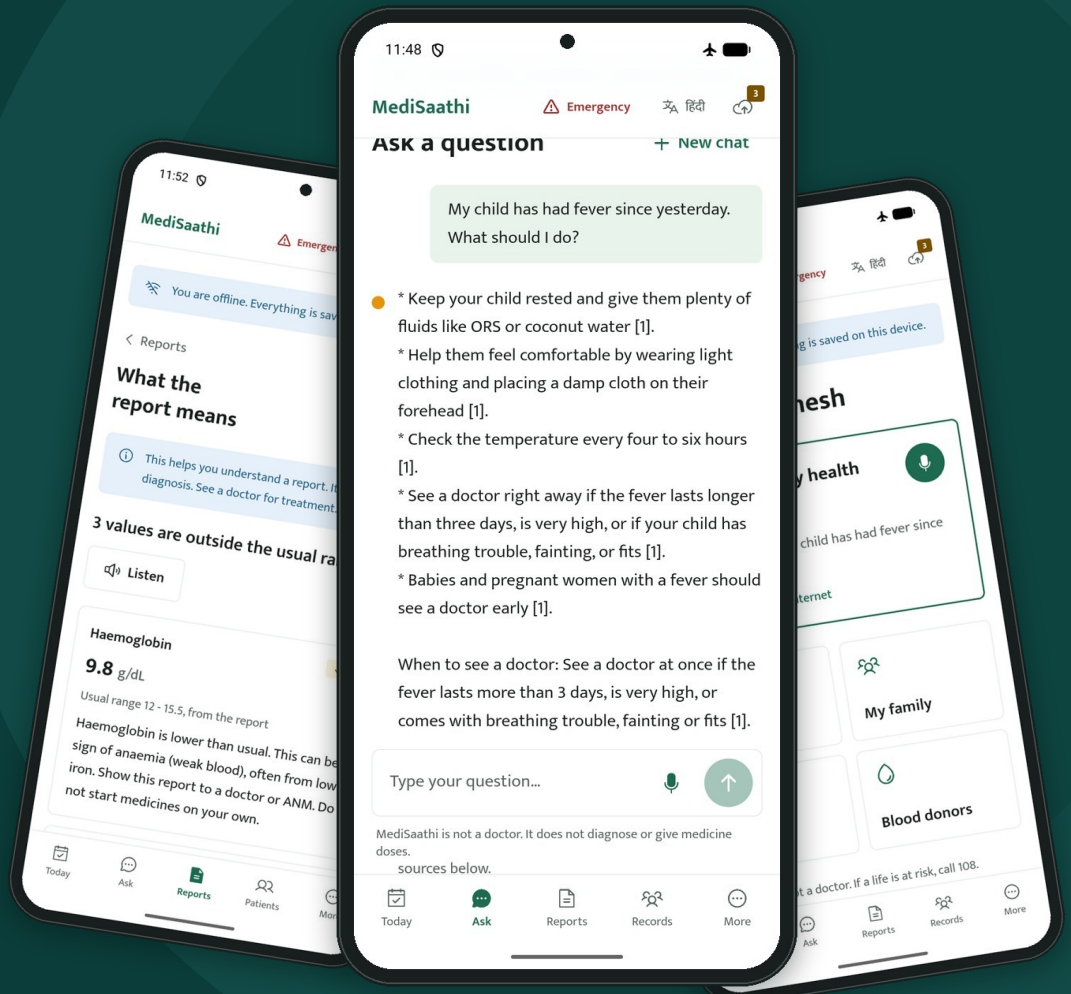
Health help that works in airplane mode.

Reads a lab report in plain words, answers health questions in multilingual Indian languages with Gemma 4 on the phone, scans medicines, tracks vaccines and outbreaks, finds blood donors. Nothing waits for a signal.

0 bars
needed for any feature

Gemma 4
runs on the phone

Multilingual
voice and text



Gemma 4 answer, no internet

TEAM

Plot Armor

PS	MY	SK	SK
Parth Sharma	**Monishka Yadav**	**Shrey Kataria**	**Shresth Kataria**
Team Leader	Team Member	Team Member	Team Member

GitHub: github.com/VenomDevX/MediSathi

Demo video: YouTube

Healthcare apps fail exactly where care is needed

No signal, no help
When the network drops, cloud apps stop working in villages and relief camps.

Reports are hard to read
“Hb 9.8 g/dL” means little to most families, so worry or wrong steps follow.

Records are fragmented
Paper registers get wet or lost, and follow-up visits quietly slip.

Donor info goes stale
In an emergency, nobody knows who nearby can give blood.

	Cloud app	MediSaathi
Explain a report	✕ Online only	✓ **On device**
Patient records	✕ Online only	✓ **Local first**
Donor search	✕ Online only	✓ **Cached directory**
First-aid guidance	✕ Online only	✓ **Reviewed library**
Health answers	✕ Unverified AI	✓ **Rules + sources**

WHO WE SERVE Four users, one offline toolkit

Patient / caregiver
- Report explained simply
- Saved reports offline
- Knows when to seek care

ASHA worker
10 lakh+ in India

- Registers patients in the field
- Offline follow-up list
- Secure sync on reconnect

Relief worker
- Fast, reviewed first aid
- Cached donors and records
- Low power and storage

Supervisor / admin
- Manages staff access
- Reviews sync failures
- Runs donor consent

Four modules, one sync layer, zero bars needed

MediScan
Report OCR + plain language

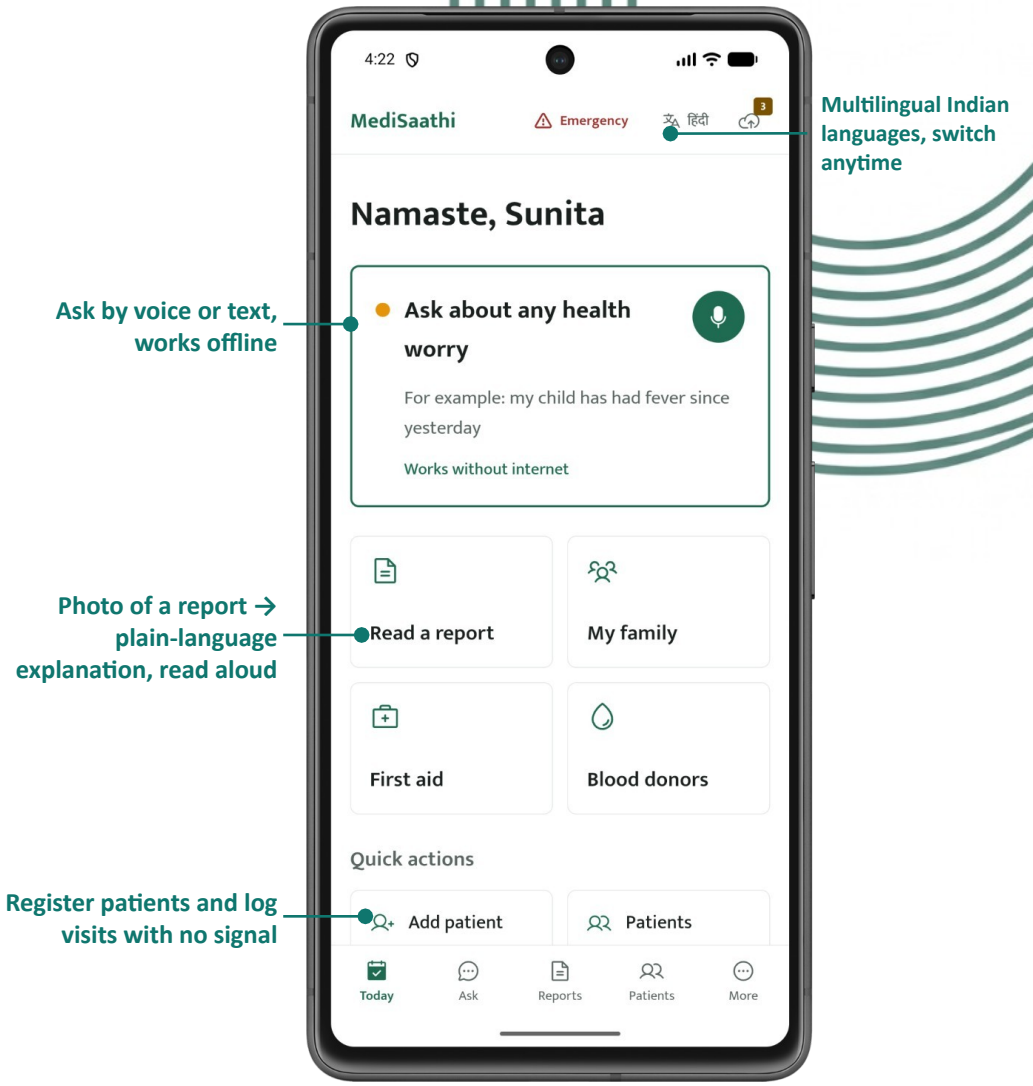
ASHA Assist
Patients, visits, follow-ups

Local DB
on device

BloodLink
Cached donor search

Health Guidance
Reviewed first-aid library

Sync service → FastAPI + PostgreSQL, when online



DESIGN PRINCIPLES Safe by design, not by prompt

Offline first
Essential workflows never wait for a server.

Privacy by default
Consent, encryption and scoped access built in.

Rules before AI
Reviewed logic decides; AI only rewords.

Humans decide
Diagnosis and treatment stay with clinicians.

Ask in your language. Gemma rewords; it never decides.

Question
Multilingual, voice or text

→ **Red flag?**

Yes → **Call 108 card**
Snake bite, not breathing, heavy bleeding: first steps, model never asked

No →

1 Library search
Reviewed first-aid cards, Hindi + English (SQLite FTS)

→ **2 Gemma 4 E2B**
On the phone via Google LiteRT-LM, uses only the matched card

→ **3 Guard**
Needs a citation, no new numbers, no medicine names

→ **4 Answer + source**
Every answer shows where it came from

↪ Guard fails? The reviewed card is shown as is.

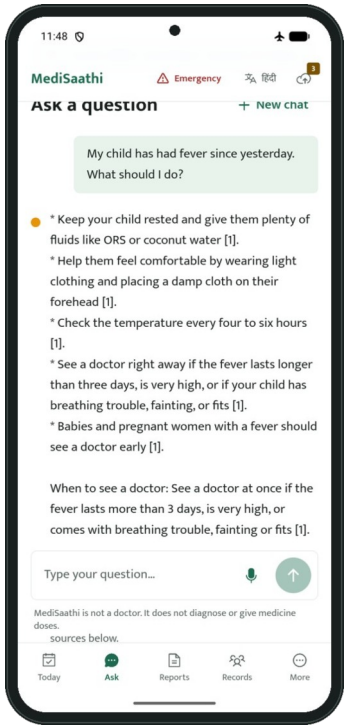
Safe by design: rules decide, Gemma only rewords, a guard checks every answer.

Reviewed first-aid library, searchable offline

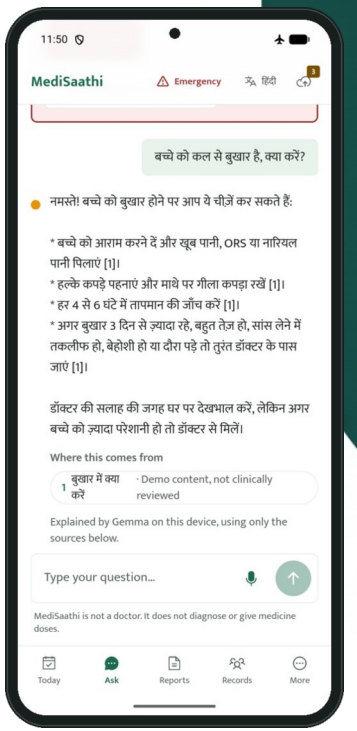
- Bleeding and wounds
- Burns
- Choking
- Diarrhoea and dehydration
- Dizziness or fainting
- Dog or animal bite
- Heat stroke

Emergency numbers and danger signs are on every screen, behind one tap: no model, no account, no network.

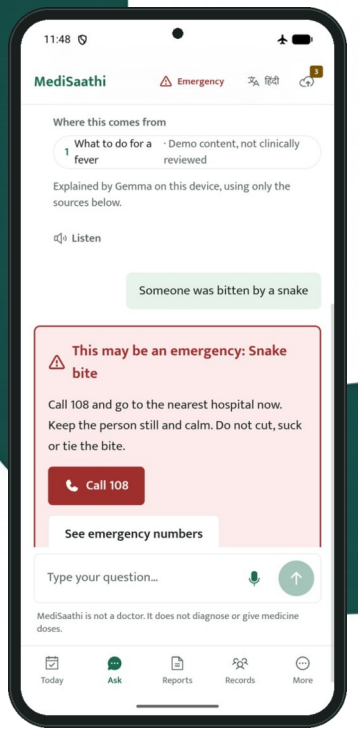
English



हिंदी



Emergency: AI skipped



108
Ambulance
Free, across India

102
Mother and child
Transport to hospital

112
National emergency
Police, fire, ambulance

Three modules working with no signal today

MODULE 01

MediScan

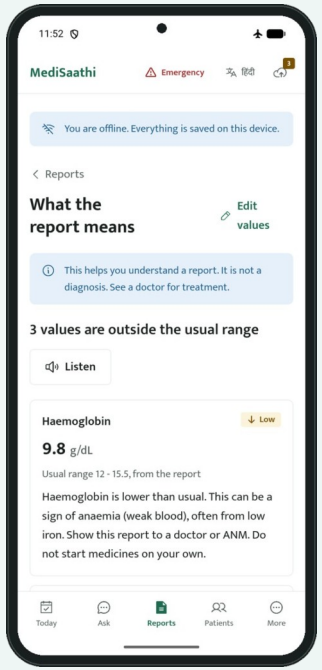


Photo or PDF of the report

Text read on the phone (Tesseract, Hindi + English)

Unclear values flagged for you to fix

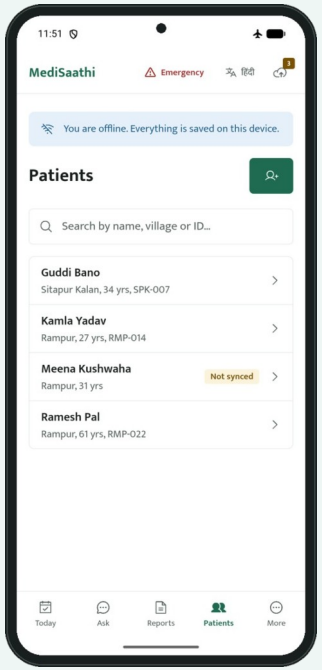
Range check: low, normal or high

Reviewed explanation, read aloud

Unknown tests are not guessed (“please ask your doctor”). Ranges come from the report; a fallback range is labelled. Not a diagnosis: said on every report.

MODULE 02

ASHA Assist

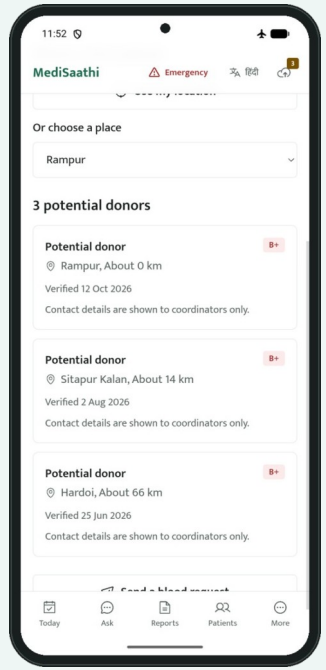


- Register patients and log visits with no signal
- Everything saved on the phone first, then synced
- “Not synced” badge until the server confirms
- PIN unlock; works up to 7 days offline

Each worker only syncs their own villages; the server checks every record.

MODULE 03

BloodLink



- Same blood group only, nearest first
- Only town and distance shown, never an exact location
- “Potential donor”: the blood bank confirms
- Cached directory works offline

A donor’s number is shared only after they say yes.

One React app on web, PWA and Android, safe on a lost phone

On the device: everything works offline

Screens	React 19, Tailwind v4, Multilingual Indian languages, voice in and out
Answers	Emergency rules → library search (SQLite FTS) → Gemma 4 → guard
Data	SQLite on the device: patients, visits, reports, sync queue
Reports	Tesseract OCR (Hindi + English) and pdf.js, on the device

Browser / PWA
Gemma via LiteRT-LM on WebGPU · sqlite-wasm on OPFS, service worker

Android (Capacitor 8)
Gemma via native LiteRT-LM, GPU then CPU · same code, same database schema

HTTPS when online →

FastAPI server

- Sign up or Google sign-in; Argon2, rotating tokens
- Idempotent push, scoped pull
- BloodLink requests with donor consent
- Optional Gemini explainer; the key stays on the server
- SQLAdmin panel, PostgreSQL in production

React + TypeScript + Vite · LiteRT-LM + Gemma 4 E2B · Capacitor 8 · FastAPI + PostgreSQL · Playwright + Vitest + pytest

Safe on a lost or shared phone

PIN unlock, offline
PBKDF2 with 310,000 rounds. Five wrong PINs lock the account. Offline use ends after 14 days.

Tokens sealed by the PIN
Refresh token AES-GCM encrypted with a PIN-derived key, never stored in the clear.

Sign up or Google, safely
Google tokens checked with Google; new accounts are citizens only. Failed tries limited.

Server decides access
Every synced record checked against the worker's area. Other users' operations refused.

Donor consent first
A donor's number is shared only after they accept a request. Locations are coarse.

Nothing leaks by backup
Android cloud backup and device transfer off. Secrets and keys kept out of git.

The app never

- ✕ diagnoses
- ✕ gives medicine doses
- ✕ shows a donor's number without consent
- ✕ uses real patient data in the demo

Sync that can't duplicate or lose a visit

1 Save	2 Queue	3 Push	4 Replay safe	5 Confirm	6 Synced
Record + op, one transaction	**Unique operation_id · pending**	**POST /sync/push · backoff**	**Replay? Return same result**	**Ack: status · version**	**Marked synced only after ack**
Change and its operation id written together	Badge shows “Not synced”	Sent when a connection appears, with retry	A repeated operation id returns the first result	Server answers with status and version	Nothing is marked done until the server confirms
Device (local DB)	Sync queue		FastAPI server		Device (local DB)

Proved by tests, not promised

- ✓ Sending the same operation twice changes nothing
- ✓ Another user's operation id is refused, not overwritten
- ✓ Records outside the worker's assigned area are rejected
- ✓ A pull returns only records from the worker's own area

Always visible on the phone

Not synced | Syncing… | Will retry | Synced

The header’s cloud badge counts what is still only on this phone; failed syncs retry on their own.

More features, all offline

1. Hands-free Saathi: speak, hear the answer
2. Medicine strip scan: use, warning, expiry
3. Outbreak alert on an offline village map
4. Vaccines and pregnancy checkups due
5. Nearest PHC and 24-hour hospital
6. SMS to donors, one message each
7. ABHA number + ABDM FHIR record export
8. DigiLocker ABHA check (demo)

Future Scope : real ABDM and DigiLocker linking, mesh networking, embeddings, advanced conflict UI

Built, tested and ready for a field pilot

211
unit tests
guard, sync, schedules, FHIR

84
backend tests
sign-up, Google, scope, visit checks

34 × 2
end-to-end flows
on phone and laptop sizes

2.6 GB
Gemma 4 E2B on Android
2.0 GB build in the browser

30 s
full Gemma answer
emulator CPU, no GPU, airplane mode

Multilingual
Indian languages
Hindi and English read aloud

Next: five gates

Sequence shown, not durations: scope and device tiers lock only after Phase 0 measurements.

GATE 01
Technical PoC
Gemma 4 + sync running offline

GATE 02
Functional MVP
Measure on real phones: GPU speed on low-cost Android

GATE 03
Safety & Security
Clinical review: library and report text checked by doctors

GATE 04
Controlled pilot
Field pilot: ASHA workers in two villages

GATE 05
Production
ABDM + DigiLocker linking, after sandbox approval

Limitations, and our plan

- **68 s per answer on emulator CPU**
  → measure GPU speed on low-cost phones
- **2.6 GB model on Android**
  → device tiers set after Phase 0
- **Library is demo content, not yet clinically reviewed**
  → doctor review at Gate 03
- **OCR can misread a photo**
  → you confirm flagged values first
- **Health library in Multilingual language**
  → other languages after clinical review

Health support that works when the network doesn't.

Decision support, not diagnosis. Clinicians stay responsible for treatment and transfusion decisions.

Village health programmes | Disaster and relief camps | Blood-bank coordination | ABHA-linked records

Try the demo

Sunita Devi ASHA worker: patients, visits, sync

Ramesh Kumar Citizen: reports, family, Ask

PIN 1234, or sign up with Google. Airplane mode on.

GitHub repo Demo video on YouTube

How MediSaathi answers each judging criterion

1

Innovation & Creativity

Gemma 4 runs on the phone with zero bars. Rules decide, the AI only rewords, and a guard checks every answer.

Slides 3 and 4

2

Relevance to the Problem

Built for where cloud apps fail: villages, relief camps and emergencies with no signal.

Slide 2

3

Technical Implementation

One React codebase on web, PWA and Android. Sync that cannot duplicate or lose a visit.

Slides 6 and 7

4

UI/UX & Presentation

Voice or text in Indian languages, reports read aloud, emergency help one tap away.

Slides 4 and 5

5

Real World Impact

Serves families, 10.29 lakh ASHA workers and blood donors who give consent first.

Slides 2 and 10

6

Scalability & Feasibility

Runs on phones people already own. Five gates from proof of concept to ABDM linking.

Slide 8

7

Team & Communication

Four members, a live demo in airplane mode, and limitations stated upfront.

Slides 1 and 8

8

Hackathon themes

Healthcare & MedTech, AI & ML, and social impact, in one offline app.

Every slide

What we learned before we wrote a line of code

What we found and what it made us build

48 vs 127

[1] Internet connections per 100 people: rural vs urban India

So every feature works with zero bars.

10.29 lakh

[2] ASHA workers, each caring for about 1,000 people

So her records live on her phone and sync later.

119 of 195

[3] Countries lack enough blood, including all of South Asia

So BloodLink shows nearby donors, with consent.

AI assists

[4] WHO: people, not algorithms, stay in charge of care decisions

So rules decide, and Gemma only rewords.

References

1 **TRAI.** Indian Telecom Services Performance Indicators, Jan–Mar 2026. trai.gov.in

2 **National Health Mission, MoHFW.** ASHA programme: 10.29 lakh ASHAs, June 2024. nhm.gov.in

3 **Roberts N. et al.** The global need and availability of blood products. Lancet Haematology, 2019.

4 **World Health Organization.** Ethics and governance of artificial intelligence for health, 2021.

5 **Google DeepMind.** Gemma 4 model card; LiteRT-LM; Google Identity Services. ai.google.dev

6 **Government of India.** Digital Personal Data Protection Act, 2023.

7 **NHM and MHA.** 108 / 102 ambulance services and ERSS 112 emergency number.

8 **Open source:** Tesseract OCR, pdf.js, SQLite WASM, Capacitor, FastAPI.

Our own tests: a Pixel 7 in airplane mode, 68 s per Gemma answer on CPU; 211 unit, 84 backend and 68 end-to-end checks.